RQ Answers

omitted

2024-04-05

Data overview

```
glimpse(all_data)
```

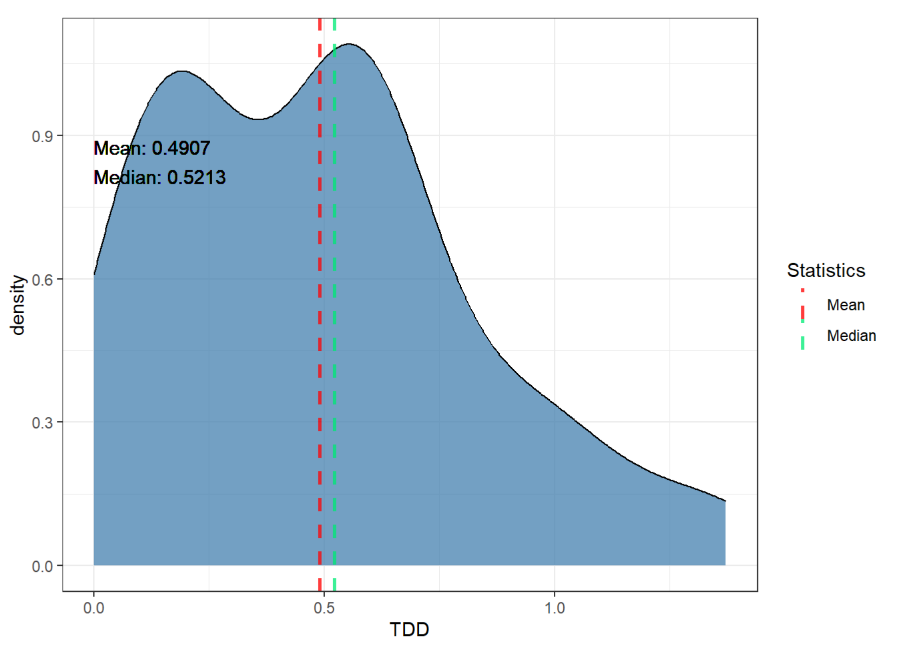
```
## Rows: 3,400
## Columns: 11
## $ repo                <chr> "httpcomponents-client", "httpcomponents-client", …
## $ pr_number           <int> 129, 141, 141, 141, 141, 141, 141, 141, 141, 141, …
## $ rule                <chr> "java:S2129", "java:S1192", "java:S1192", "java:S2…
## $ severity            <chr> "MAJOR", "CRITICAL", "CRITICAL", "MAJOR", "MAJOR",…
## $ file                <chr> "httpclient5/src/test/java/org/apache/hc/client5/h…
## $ type                <chr> "CODE_SMELL", "CODE_SMELL", "CODE_SMELL", "CODE_SM…
## $ status              <chr> "OPEN", "OPEN", "OPEN", "OPEN", "OPEN", "OPEN", "O…
## $ debt                <int> 5, 28, 20, 2, 2, 2, 2, 10, 8, 2, 2, 2, 2, 2, 2, 2,…
## $ ncloc_affected_file <int> 73, 585, 585, 585, 585, 585, 585, 585, 585, 585, 5…
## $ complexity          <int> 4, 55, 55, 55, 55, 55, 55, 55, 55, 55, 55, 55, 55,…
## $ origin              <chr> "PRE-EXISTING", "PRE-EXISTING", "PRE-EXISTING", "P…
```

RQ1

TDD (Technical Debt Density) by PR Distribution

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdd), : All aesthetics have length 1, but the da
ta has 48 rows.
## ℹ Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdd), : All aesthetics have length 1, but th
e data has 48 rows.
## ℹ Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```

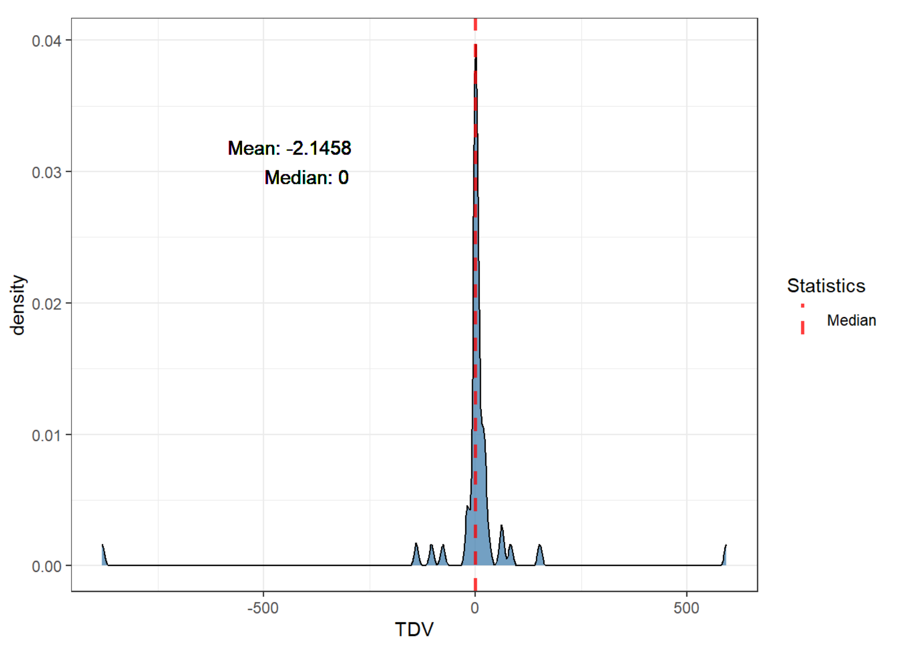


TDV (Technical Debt Variation)

TDV Distribution

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdv), : All aesthetics have length 1, but the da
ta has 48 rows.
## ℹ Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdv), : All aesthetics have length 1, but th
e data has 48 rows.
## ℹ Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```



Percentages for TDV < 0 (TD decrease), TDV = 0 (TD unchanged) and TDV > 0 (TD increased) by repo

```
## # A tibble: 1 × 4
##   repo                 tdv_lower_than_zero tdv_equal_to_zero tdv_greater_than_zero
##   <chr>                              <dbl>             <dbl>                 <dbl>
## 1 httpcomponents-cl…                  16.7              43.8                  39.6
```

Mean of percentages of TDV (mean of repo percentages)

```
## # A tibble: 1 × 3
##   mean_tdv_lower_than_zero mean_tdv_equal_to_zero mean_tdv_greater_than_zero
##                      <dbl>                  <dbl>                      <dbl>
## 1                     16.7                   43.8                       39.6
```

Percentages of pre-existing TDV by repo

```
## # A tibble: 1 × 3
##   repo                  preexisting_tdv_equal_to_zero preexisting_tdv_greater_…¹
##   <chr>                                         <dbl>                      <dbl>
## 1 httpcomponents-client                          64.4                       35.6
## # ℹ abbreviated name: ¹preexisting_tdv_greater_than_zero
```

Mean of percentages of pre-existing TDV (mean of repo percentages)

```
## # A tibble: 1 × 2
##   preexisting_tdv_equal_to_zero mean_preexisting_tdv_greater_than_zero
##                           <dbl>                                  <dbl>
## 1                          64.4                                   35.6
```

Percentages for new TDV by repo

```
## # A tibble: 1 × 3
## # Groups:   repo [1]
##   repo                  new_tdv_equal_to_zero new_tdv_greater_than_zero
##   <chr>                                 <dbl>                     <dbl>
## 1 httpcomponents-client                  95.6                      4.35
```

Mean of percentages of new TDV (mean of repo percentages)

```
## # A tibble: 1 × 2
##   mean_new_tdv_equal_to_zero mean_new_tdv_greater_than_zero
##                        <dbl>                          <dbl>
## 1                       95.6                           4.35
```

RQ2

As data is unbalanced across repositories, as shown in the issue characterization (../issues_characterization.Rmd), we will examine the top 10 fixed issues and the top 10 unfixed issues for each repository, and then aggregate them into a rank using the $PositionScore$ metric given by the following formula:

$$PositionScore_i = \sum_{j=1}^{k} Position_j$$

given a rule i, its $PositionScore_i$ will be the sum of its position across all k repositiries.After calculating the metrics for all rules, we rank them in a TOP 10 of the most frequent rules (fixed or unfixed, depending on the context) across repositories. The lower the PositionScore, the more frequent the rule across repositories.

TOP 10 fixed per repo ranked by count

All top 10 fixed rules for each repo.

```
## # A tibble: 10 × 2
## # Groups:   rank [10]
##     rank `httpcomponents-client`
##    <int> <chr>
##  1     1 java:S1192  -  84
##  2     2 java:S1874  -  8
##  3     3 java:S1117  -  8
##  4     4 java:S100  -  7
##  5     5 java:S112  -  5
##  6     6 java:S125  -  5
##  7     7 java:S1481  -  4
##  8     8 java:S1130  -  3
##  9     9 java:S1066  -  3
## 10    10 java:S2119  -  3
```

TOP 10 fixed

TOP 10 fixed rules by PositionScore metric.

```
## # A tibble: 10 × 2
##    rule       position_score
##    <chr>               <dbl>
##  1 java:S1192            122
##  2 java:S1117            123
##  3 java:S1874            124
##  4 java:S100             125
##  5 java:S112             126
##  6 java:S125             127
##  7 java:S1481            128
##  8 java:S1066            129
##  9 java:S1130            130
## 10 java:S1854            131
```

TOP 10 unfixed per repo ranked by count

All top 10 unfixed rules for each repo.

```
## # A tibble: 10 × 2
## # Groups:   rank [10]
##     rank `httpcomponents-client`
##    <int> <chr>
##  1     1 java:S1192  -  1789
##  2     2 java:S100  -  196
##  3     3 java:S106  -  194
##  4     4 java:S1124  -  130
##  5     5 java:S1075  -  122
##  6     6 java:S1117  -  79
##  7     7 java:S2119  -  61
##  8     8 java:S2259  -  60
##  9     9 java:S112  -  53
## 10    10 java:S108  -  43
```

TOP 10 unfixed

TOP 10 unfixed rules by PositionScore metric.

```
## # A tibble: 10 × 2
##    rule       position_score
##    <chr>               <dbl>
##  1 java:S1192            122
##  2 java:S100             123
##  3 java:S106             124
##  4 java:S1124            125
##  5 java:S1075            126
##  6 java:S1117            127
##  7 java:S2119            128
##  8 java:S2259            129
##  9 java:S112             130
## 10 java:S108             131
```

Outliers

Higher TDD

```
## # A tibble: 48 × 5
##    repo                  pr_number total_debt total_ncloc   tdd
##    <chr>                     <int>      <int>       <int> <dbl>
##  1 httpcomponents-client       222        354         258 1.37
##  2 httpcomponents-client       302       1073         816 1.31
##  3 httpcomponents-client       362        124         109 1.14
##  4 httpcomponents-client       529        739         702 1.05
##  5 httpcomponents-client       492        132         133 0.992
##  6 httpcomponents-client       413         35          38 0.921
##  7 httpcomponents-client       484       3724        4090 0.911
##  8 httpcomponents-client       426        716         797 0.898
##  9 httpcomponents-client       417        530         719 0.737
## 10 httpcomponents-client       421       3297        4659 0.708
## # ℹ 38 more rows
```

Higher TDV

```
## # A tibble: 48 × 3
##    pr_number repo                    tdv
##        <int> <chr>                 <int>
##  1       483 httpcomponents-client   593
##  2       543 httpcomponents-client   152
##  3       501 httpcomponents-client    83
##  4       192 httpcomponents-client    64
##  5       504 httpcomponents-client    60
##  6       496 httpcomponents-client    32
##  7       506 httpcomponents-client    23
##  8       356 httpcomponents-client    22
##  9       435 httpcomponents-client    22
## 10       537 httpcomponents-client    20
## # ℹ 38 more rows
```

Lower TDV

```
## # A tibble: 48 × 3
##    pr_number repo                    tdv
##        <int> <chr>                 <int>
##  1       484 httpcomponents-client  -882
##  2       426 httpcomponents-client  -140
##  3       454 httpcomponents-client  -104
##  4       327 httpcomponents-client   -77
##  5       362 httpcomponents-client   -20
##  6       494 httpcomponents-client   -20
##  7       544 httpcomponents-client   -15
##  8       413 httpcomponents-client    -5
##  9       129 httpcomponents-client     0
## 10       141 httpcomponents-client     0
## # ℹ 38 more rows
```

Extra

Number of PRs with pre-existing TD

```
##    n
## 1 45
```

Number of java:S1192 issues that affect test code

```
##      n
## 1 1873
```

Number of java:S1192 issues that affect not test code

```
##   n
## 1 0
```

Summary NCLOC

```
##      ncloc
##  Min.   :   34.0
##  1st Qu.:  177.2
##  Median :  408.5
##  Mean   : 1906.9
##  3rd Qu.: 1982.0
##  Max.   :25976.0
```